

CENG381 Stochastic Processes

Renewal Processes - Answer Key 1

Q1 Solution

a) Distribution of $N(t)$:

When inter-renewal times are Exponential(λ), the renewal process is a POISSON PROCESS with rate λ .

$$N(t) \sim \text{Poisson}(\lambda * t)$$

$$E[N(t)] = \lambda * t$$

This follows because the Poisson process is the unique renewal process with exponentially distributed inter-arrival times (memoryless property).

b) Long-run rate and expected count over 40 hours:

Elementary Renewal Theorem:

$$\text{Long-run rate} = 1 / E[X] = 1 / (1/\lambda) = \lambda = 0.5 \text{ per hour}$$

Expected number of replacements in 40 hours:

$$E[N(40)] = \lambda * 40 = 0.5 * 40 = 20 \text{ bulbs}$$

Interpretation: on average, one bulb is used every 2 hours, so 20 bulbs are used in a 40-hour period.

c) $P(N(4) \geq 3)$ with $\lambda * t = 2$:

$$N(4) \sim \text{Poisson}(\lambda * 4) = \text{Poisson}(0.5 * 4) = \text{Poisson}(2)$$

$$P(N(4) \geq 3) = 1 - P(N(4) = 0) - P(N(4) = 1) - P(N(4) = 2)$$

$$P(N(4) = k) = e^{(-2)} * 2^k / k!$$

$$P(N=0) = e^{(-2)} * 1 / 1 = e^{(-2)} \sim 0.1353$$

$$P(N=1) = e^{(-2)} * 2 / 1 = 2 * e^{(-2)} \sim 0.2707$$

$$P(N=2) = e^{(-2)} * 4 / 2 = 2 * e^{(-2)} \sim 0.2707$$

$$P(N(4) \geq 3) = 1 - e^{(-2)} - 2 * e^{(-2)} - 2 * e^{(-2)}$$

$$= 1 - 5 * e^{(-2)}$$

$$\sim 1 - 5 * 0.1353 = 1 - 0.6767 = 0.3233$$

CENG381 Stochastic Processes

Renewal Processes - Answer Key 1

About a 32.3% chance of 3 or more replacements in the first 4 hours.

Q2 Solution

a) $E[X]$ and $E[X^2]$ for $X \sim \text{Uniform}[1,3]$:

$$E[X] = (1 + 3)/2 = 4/2 = 2 \text{ minutes}$$

$$\begin{aligned} E[X^2] &= (1^2 + 1 \cdot 3 + 3^2) / 3 \\ &= (1 + 3 + 9) / 3 \\ &= 13/3 \sim 4.333 \text{ minutes}^2 \end{aligned}$$

Alternative derivation via integral:

$$\begin{aligned} E[X^2] &= \text{integral from 1 to 3 of } x^2 * (1/2) dx \\ &= (1/2) * [x^3/3] \text{ from 1 to 3} \\ &= (1/2) * (27/3 - 1/3) \\ &= (1/2) * (26/3) \\ &= 13/3 \text{ (OK)} \end{aligned}$$

b) Long-run rate and 60-minute count:

$$\text{Long-run rate} = 1 / E[X] = 1/2 \text{ bus per minute} = 0.5 \text{ buses/min}$$

$$\text{Expected buses in 60 minutes} = 60 * (1/2) = 30 \text{ buses}$$

c) $m(1.5)$ for $\text{Uniform}[1,3]$:

For $1 \leq t < 2$, the first bus cannot arrive before $t=1$ (since $X_1 \geq 1$), and a second bus cannot arrive before $t = 1 + 1 = 2$ (since $X_1 + X_2 \geq 2$). So for $1 \leq t < 2$, $N(t)$ is either 0 (no buses yet) or 1 (first bus arrived).

$$m(t) = E[N(t)] = P(N(t) \geq 1) = P(X_1 \leq t) \quad \text{for } 1 \leq t < 2$$

$$m(1.5) = P(X_1 \leq 1.5) = (1.5 - 1) / (3 - 1) = 0.5 / 2 = 0.25$$

So on average, 0.25 buses have arrived by time $t = 1.5$ minutes. This makes sense: there's a 25% chance the first bus arrived by 1.5 min, and a 75% chance it hasn't arrived yet.

CENG381 Stochastic Processes

Renewal Processes - Answer Key 1

Q3 Solution

a) $E[X]$ and $E[X^2]$:

$$\begin{aligned}E[X] &= 1 \cdot 0.5 + 2 \cdot 0.3 + 4 \cdot 0.2 \\&= 0.5 + 0.6 + 0.8 \\&= 1.9 \text{ seconds}\end{aligned}$$

$$\begin{aligned}E[X^2] &= 1^2 \cdot 0.5 + 2^2 \cdot 0.3 + 4^2 \cdot 0.2 \\&= 0.5 + 4 \cdot 0.3 + 16 \cdot 0.2 \\&= 0.5 + 1.2 + 3.2 \\&= 4.9 \text{ seconds}^2\end{aligned}$$

b) Long-run rate:

$$\text{Long-run rate} = 1 / E[X] = 1 / 1.9 \sim 0.526 \text{ completions per second}$$

About 0.526 batch completions per second, or roughly one every 1.9 seconds.

c) $P(N(3) \geq 1)$ and $P(N(3) \geq 2)$:

$$\begin{aligned}\text{(i) } P(N(3) \geq 1) &= P(S_1 \leq 3) = P(X_1 \leq 3) \\&= P(X=1) + P(X=2) \quad [\text{since } P(X=4) \text{ corresponds to } X=4 > 3] \\&= 0.5 + 0.3 = 0.8\end{aligned}$$

$$\text{(ii) } P(N(3) \geq 2) = P(S_2 \leq 3) = P(X_1 + X_2 \leq 3)$$

Enumerate pairs (X_1, X_2) with values in $\{1, 2, 4\}$ and sum ≤ 3 :

- (1, 1): sum = 2 $\leq$ 3, prob = $0.5 \cdot 0.5 = 0.25$
- (1, 2): sum = 3 $\leq$ 3, prob = $0.5 \cdot 0.3 = 0.15$
- (2, 1): sum = 3 $\leq$ 3, prob = $0.3 \cdot 0.5 = 0.15$
- (1, 4): sum = 5 $>$ 3, excluded
- (2, 2): sum = 4 $>$ 3, excluded
- (4, *): sum $\geq$ 5 $>$ 3, excluded

$$P(S_2 \leq 3) = 0.25 + 0.15 + 0.15 = 0.55$$

Interpretation: there is a 55% chance that at least 2 batches complete within the first 3 seconds.